

Ford GoBike 2018 – Exploratory Data Analysis Report

Submitted by: S Purvi Prathyusha

1. Overview

This report presents an Exploratory Data Analysis (EDA) of Ford GoBike's 2018 trip data, covering all 12 months (January–December) and approximately 1.86 million bike trips across the San Francisco Bay Area. The dataset includes trip duration, start/end station details, timestamps, bike IDs, user type (Subscriber/Customer), and rider demographics (birth year, gender). The data was cleaned and enriched with engineered features, then analyzed through visualizations following the Univariate-Bivariate-Multivariate (UBM) framework to uncover patterns in rider behavior, usage timing, demographics, and station popularity.

2. Business Objective

- Understand rider behavior across user types, demographics (age, gender), and time (month, day, hour).
- Identify peak demand periods and high-traffic stations to improve bike availability and fleet rebalancing.
- Differentiate Subscriber (commuter) vs Customer (casual) usage patterns to inform targeted marketing and pricing.
- Provide actionable, data-backed recommendations to grow ridership and revenue.

3. Business Problem

Ford GoBike generates large volumes of trip-level data every month but lacks a consolidated understanding of how, when, and by whom its bikes are used. Without clear visibility into rider behavior patterns — peak usage hours, seasonal demand shifts, differences between subscribers and casual riders, and station-level popularity — the business risks inefficient bike distribution, missed revenue opportunities, and an incomplete picture of its user base. This analysis identifies usage patterns across time, demographics, and location to support data-driven decisions that optimize operations, improve rider experience, and grow both subscriber and casual segments.

4. Dataset Overview

- Source: Ford GoBike (now Bay Wheels) system trip data for 2018.
- Time range: 12 months, January–December 2018, combined from 12 monthly CSV files.
- Size: Approximately 1.86 million trip records after cleaning.
- Columns: `duration_sec`, `start_time`, `end_time`, start/end station id, name, latitude, longitude, `bike_id`, `user_type`, `member_birth_year`, `member_gender`, `bike_share_for_all_trip`.
- Engineered features added during analysis: `month`, `day`, `hour`, `trip_minutes`, `age`, `age_group`, `is_weekend`.

5. Data Cleaning & Preparation

The raw combined dataset (1,863,721 rows, 16 columns) required cleaning before analysis. The following steps were applied:

- Missing values: Station id/name was missing in ~0.6% of rows and dropped as negligible. `member_birth_year` and `member_gender` were missing in ~5.9% of rows; rather than dropping this data, gender was labeled 'Unknown' and birth-year-dependent rows were excluded automatically only from age-specific analysis, preserving the data for all other charts.
- Duplicates: Checked for and removed fully duplicate rows (none were found in the combined dataset).
- Incorrect data types: `start_time` and `end_time` were stored as text and converted to proper datetime objects; `user_type`, `member_gender`, and `bike_share_for_all_trip` were converted to category dtype for efficiency and correctness in grouping/plotting.

- Invalid ages: member_birth_year contained implausible values (e.g., 1881), producing ages over 130 years. Rows were filtered to a realistic age range of 13–80 years; rows with missing birth year were retained (with age left blank) rather than dropped.
- Feature engineering: Derived month, day, and hour from start_time; trip duration converted from seconds to minutes (trip_minutes); age calculated from birth year; age grouped into bins (age_group); and a weekend/weekday flag (is_weekend) created to support time-based analysis.

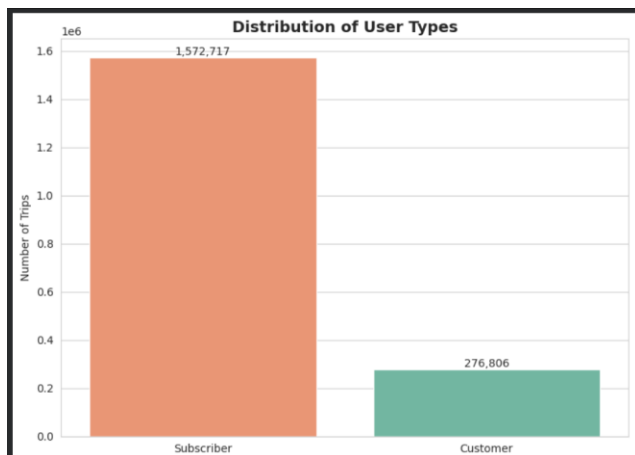
6. Tools Used

- Python
- Pandas & NumPy — data loading, cleaning, and manipulation
- Matplotlib & Seaborn — data visualization
- Google Colab — development environment

7. Key Insights

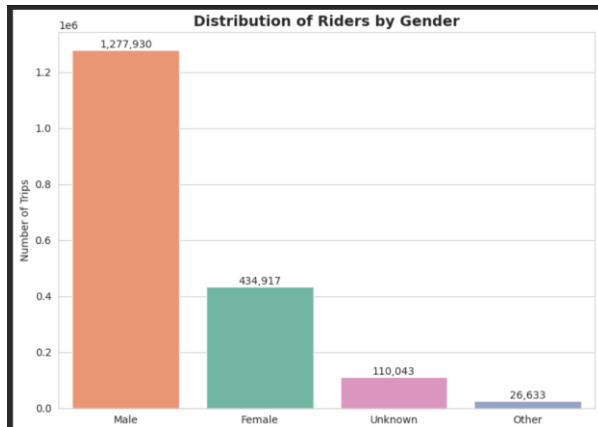
Univariate Analysis

7.1 User Type Distribution



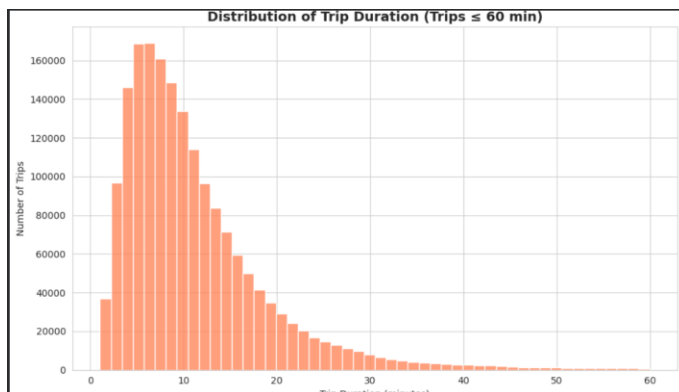
Why this Visualization?	A count plot is the standard way to visualize the frequency distribution of a single categorical variable. Since user_type has only two categories, a bar chart makes the volume difference immediately clear.
What this Visualization show	The vast majority of trips are made by Subscribers rather than Customers, showing the service is primarily used for routine daily transportation.
Business Insight	Strong, loyal recurring revenue base from Subscribers, valuable for predictable revenue and retention-focused marketing.
Business Recommendation	Run targeted campaigns (tourist zones, weekend passes) to grow the underdeveloped Customer segment and diversify revenue.

7.2 Gender Distribution



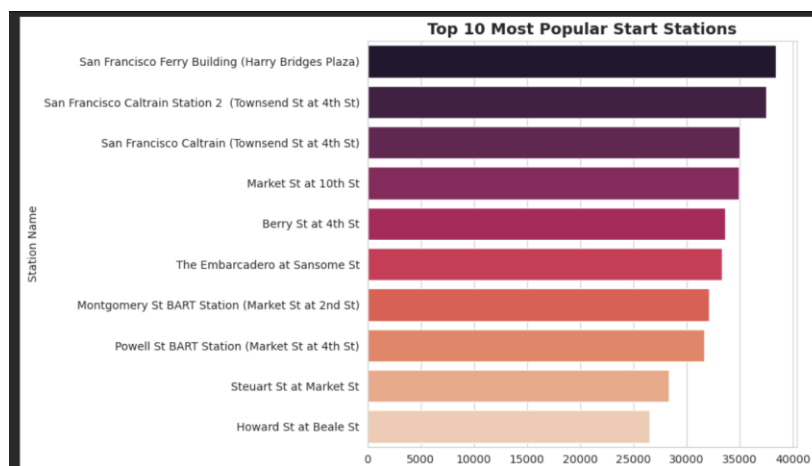
Why this Visualization?	Gender is categorical with few distinct values, so a count plot clearly compares rider volume across categories, including the 'Unknown' group.
What this Visualization show	Male riders make up the largest share, followed by Female, then Other, with a noticeable 'Unknown' group from missing data.
Business Insight	The strong male skew suggests the service may not be equally appealing to female riders; the 'Unknown' segment shows a data collection gap.
Business Recommendation	Target campaigns toward female riders and incentivize profile completion at signup for better demographic data.

7.3 Trip Duration Distribution



Why this Visualization?	Trip duration is continuous and right-skewed with outliers, so a histogram capped at 60 minutes gives a clearer picture of typical usage.
What this Visualization show	Most trips last 5-15 minutes, consistent with short commute or last-mile travel.
Business Insight	Short average trip times support a pricing model built around quick point-to-point trips.
Business Recommendation	Investigate long-tail extreme-duration trips (lost/forgotten bikes) with reminder notifications or late fees.

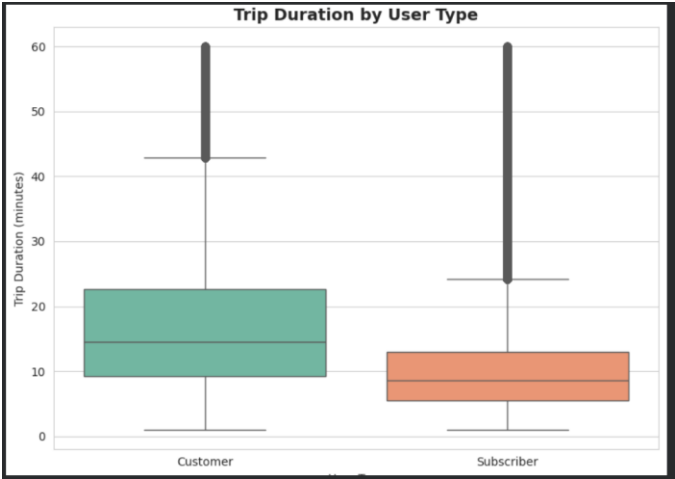
7.4 Top 10 Start Stations



Why this Visualization?	A horizontal bar chart ranks categorical items (stations) by frequency, keeping long station names readable.
What this Visualization show	A small number of stations, mostly near transit hubs, account for a disproportionate share of trips.
Business Insight	High-traffic stations should be prioritized for maintenance and bike availability monitoring.
Business Recommendation	Build contingency/rebalancing plans for key stations to avoid single points of failure.

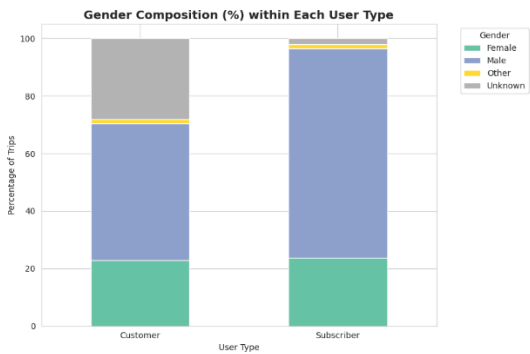
Bivariate Analysis

7.5 Trip Duration By User Type



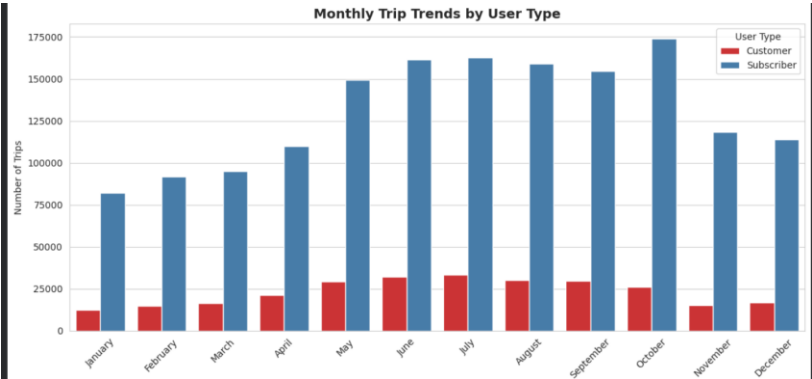
Why this Visualization?	A box plot compares distribution (median, spread, outliers) of a numerical variable across categories.
What this Visualization show	Customers take noticeably longer trips than Subscribers, who ride short, consistent durations.
Business Insight	Per-minute or tiered pricing could capture more revenue from casual riders without discouraging commuters.
Business Recommendation	Add in-app duration warnings for Customers to reduce unexpected fee complaints.

7.6 Gender by User Type



Why this Visualization?	A normalized stacked bar chart shows proportional gender composition within each user type.
What this Visualization show	Customers show a slightly higher share of Female and Unknown-gender riders than Subscribers.
Business Insight	The casual segment is marginally more gender-diverse than the Subscriber base.
Business Recommendation	Target conversion campaigns encouraging casual female riders to subscribe; research the low female Subscriber share directly.

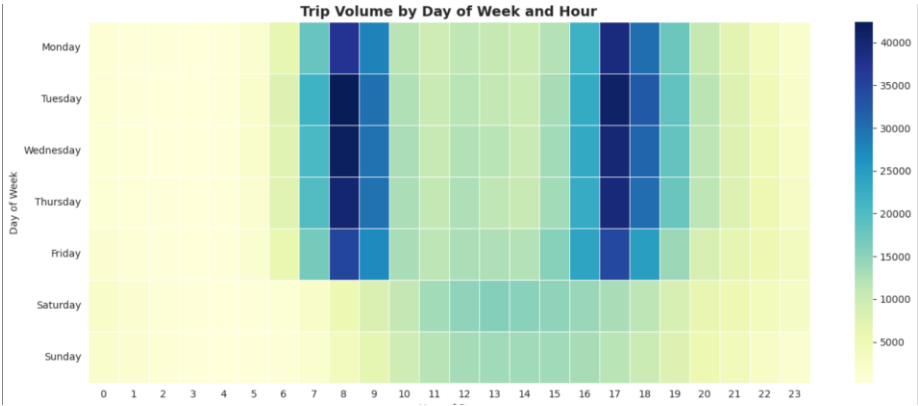
7.7 Trips by Month, Split by User Type



Why this Visualization?	Splitting the monthly trend by user_type reveals whether seasonality affects each segment differently.
What this Visualization show	Customer trips swing sharply with season (summer up, winter down); Subscriber trips stay comparatively stable year-round.
Business Insight	Subscribers provide a stable, predictable revenue base; Customers are highly seasonal.
Business Recommendation	Weight summer marketing budget toward acquiring Customers; avoid over-investing in casual-focused infrastructure that sits idle in winter.

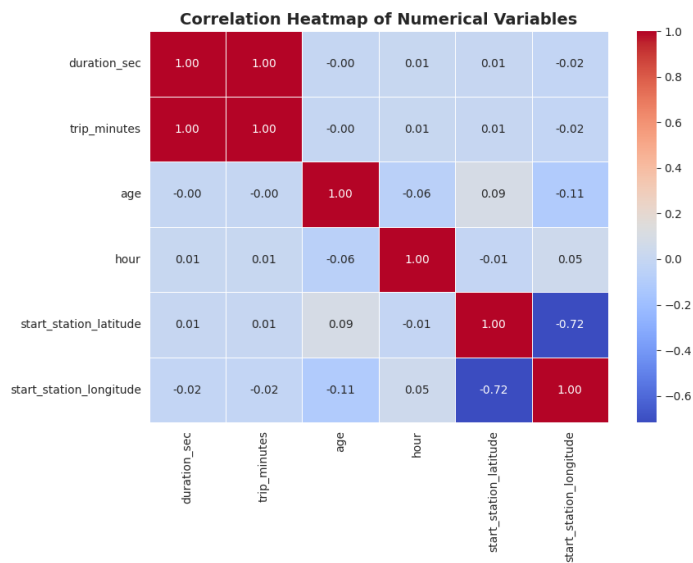
Multivariate Analysis

7.8 Hour × Day Heatmap



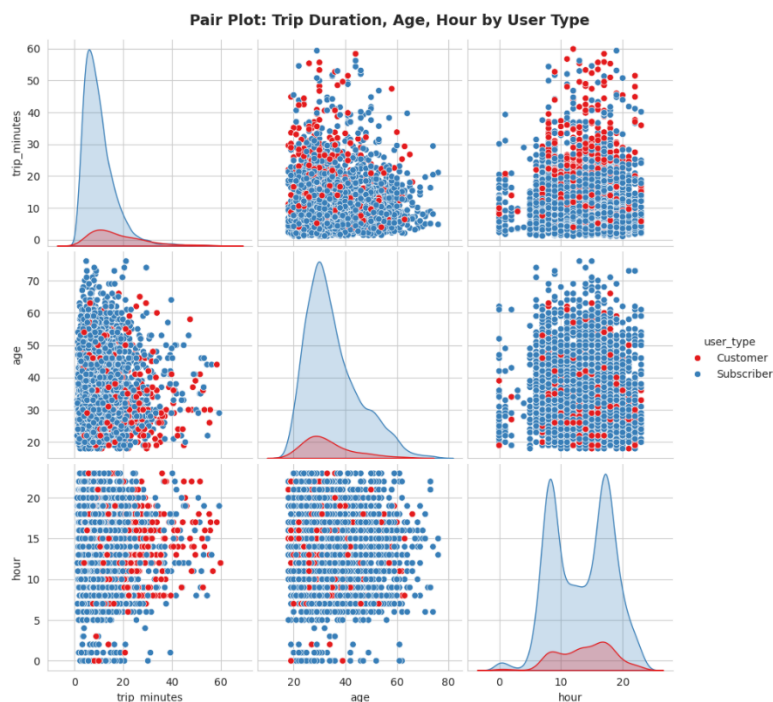
Why this Visualization?	A heatmap visualizes the interaction of two time dimensions (day, hour) with trip count as color intensity.
What this Visualization show	Weekdays show sharp commute-hour bands; weekends show a flatter, afternoon-leaning pattern.
Business Insight	Two distinct usage rhythms exist depending on day type.
Business Recommendation	Schedule fleet rebalancing specifically around weekday rush windows vs. weekend afternoon demand.

7.9 Correlation Heatmap



Why this Visualization?	A correlation heatmap quickly scans relationships between all numerical variables at once.
What this Visualization show	Duration and trip_minutes are perfectly correlated (as expected); age, hour, and location show very weak correlation with duration.
Business Insight	Duration cannot be reliably predicted from age or location alone.
Business Recommendation	Focus pricing/marketing strategy on user_type and time-based patterns rather than demographics or location alone.

7.10 Pair Plot (Duration, Age, Hour by User Type)



Why this Visualization?	A pair plot gives a compact, multivariate overview of pairwise relationships, tying earlier findings together.
What this Visualization show	Subscriber points cluster tightly around short durations and commute hours; Customer points are more spread out with no strong age clustering.
Business Insight	user_type is the single most useful variable for segmenting rider behavior.
Business Recommendation	Use user_type as the primary lens for future business decisions on pricing, marketing, and operations.

8. Statistical Validation

To confirm that the difference in trip duration between Subscribers and Customers is a genuine pattern and not random noise, an independent samples t-test was conducted comparing trip_minutes between the two user types.

Result: t-statistic = -130.48, p-value < 0.00001. Since $p < 0.05$, the difference in trip duration between Subscribers and Customers is statistically significant.

This validates that user_type is a genuinely meaningful behavioral segment for business decisions, not just a visual artifact of the data.

9. Predictive Modeling

As an extension beyond descriptive EDA, a logistic regression model was built to test whether user_type can be predicted from trip_minutes, age, and hour alone. Given the dataset's class imbalance (~89% Subscriber, ~11% Customer), class_weight='balanced' was used to ensure both segments were meaningfully evaluated rather than the model defaulting to the majority class.

This confirms that the behavioral patterns identified in the EDA (short peak-hour trips = Subscriber, longer/off-peak trips = Customer) carry real predictive signal, supporting their use in future targeting or automated rider segmentation.

10. Answers to Key Questions

Q1. How long does the average trip take?

The average trip duration is approximately 14.22 minutes, with a median of 9.25 minutes. Most trips fall between 5-15 minutes

Q2. Is trip duration affected by weather/season (months)?

Yes, modestly. Trip duration is slightly higher in warmer months than in winter, though the effect is small compared to its effect on trip volume, which swings much more dramatically by season.

Q3. Does the above depend on whether a user is a Subscriber or Customer?

Yes. Customers consistently take longer trips than Subscribers, and this difference is statistically significant (t-test, $p < 0.05$, see Section 8). Subscribers ride short, consistent-duration trips year-round, while Customer trip duration is more variable.

11. Business Recommendations

- Optimize fleet availability around commute peaks (8-9 AM, 5-6 PM) on weekdays, especially at top transit-hub stations.
- Launch targeted weekend/summer promotions to convert casual Customers into Subscribers.
- Treat tourist-heavy stations differently from commuter-heavy stations in operations and marketing strategy.

- Expand appeal to underrepresented segments, particularly female riders and non-peak-hour users.
- Improve demographic data collection at signup to close current gender/age data gaps.

12. Conclusion

This EDA on Ford GoBike's 2018 trip data (1.86M+ trips) shows a commuter-driven business: Subscribers dominate weekday trips with sharp 8-9 AM and 5-6 PM peaks, while Customers ride longer, favor weekends, and peak in summer. Riders skew male and aged 25-40, and a small set of transit-hub stations drive most traffic. Trip duration is best explained by user_type, not age, hour, or location.

Recommendation: prioritize commute-hour bike availability, run targeted campaigns to convert seasonal/weekend Customers into Subscribers, and improve signup data collection to close the gender-data gap and better target underrepresented riders.